

Overview of Mixture model

Subject: Mixture model

1.

Application of Gaussian mixture models Reynolds, D.A.; Rose, R.C. (January 1995). "Robust text-independent speaker identification using Gaussian mixture speaker models". *IEEE Transactions on Speech and Audio Processing*. 3 (1): 72–83. Bibcode:1995ITSAP...3...72R. doi:10.1109/89.365379. S2CID 7319345. Permuter, H.; Francos, J.; Jermyn, I.H. (2003). Gaussian mixture models of texture and colour for image database retrieval. *IEEE International Conference on Acoustics, Speech, and Signal Processing, 2003. Proceedings (ICASSP '03)*. doi:10.1109/ICASSP.2003.1199538. Permuter, Haim; Francos, Joseph; Jermyn, Ian (2006). "A study of Gaussian mixture models of color and texture features for image classification and segmentation" (PDF). *Pattern Recognition*. 39 (4): 695–706. Bibcode:2006PatRe...39..695P. doi:10.1016/j.patcog.2005.10.028. S2CID 8530776. Lemke, Wolfgang (2005). *Term Structure Modeling and Estimation in a State Space Framework*. Springer Verlag. ISBN 978-3-540-28342-3. Brigo, Damiano; Mercurio, Fabio (2001). Displaced and Mixture Diffusions for Analytically-Tractable Smile Models. *Mathematical Finance – Bachelier Congress 2000. Proceedings*. Springer Verlag. Brigo, Damiano; Mercurio, Fabio (June 2002). "Lognormal-mixture dynamics and calibration to market volatility smiles". *International Journal of Theoretical and Applied Finance*. 5 (4): 427. CiteSeerX 10.1.1.210.4165. doi:10.1142/S0219024902001511. Spall, J. C.; Maryak, J. L. (1992). "A feasible Bayesian estimator of quantiles for projectile accuracy from non-i.i.d. data". *Journal of the American Statistical Association*. 87 (419): 676–681. doi:10.1080/01621459.1992.10475269. JSTOR 2290205. Alexander, Carol (December 2004). "Normal mixture diffusion with uncertain volatility: Modelling short- and long-term smile effects" (PDF). *Journal of Banking & Finance*. 28 (12): 2957–80. doi:10.1016/j.jbankfin.2003.10.017. Stylianou, Yannis; Pantazis, Yannis; Calderero, Felipe; Larroy, Pedro; Severin, Francois; Schimke, Sascha; Bonal, Rolando; Matta, Federico; Valsamakis, Athanasios (2005). *GMM-Based Multimodal Biometric Verification* (PDF). Chen, J.; Adebomi, O.E.; Olusayo, O.S.; Kulesza, W. (2010). The Evaluation of the Gaussian Mixture Probability Hypothesis Density approach for multi-target tracking. *IEEE International Conference on Imaging Systems and Techniques, 2010*. doi:10.1109/IST.2010.5548541. {{cite conference}}: CS1 maint: numeric names: authors list (link)

2.

The expectation step With initial guesses for the parameters of our mixture model, "partial membership" of each data point in each constituent distribution is computed by calculating expectation values for the membership variables of each data point. That is, for each data point x_j and distribution Y_i , the membership value $y_{i,j}$ is:

3.

with new parameters $\phi \sim i$, $\mu \sim i$ $\{\displaystyle {\tilde {\phi }}_{i},{\boldsymbol {\tilde {\mu }}}_{i}\}$ and $\Sigma \sim i$ $\{\displaystyle {\boldsymbol {\tilde {\Sigma }}}_{i}\}$ that are updated using the EM algorithm. Although EM-based parameter updates are well-established, providing the initial estimates for these parameters is currently an area of active research. Note that this formulation yields a closed-form solution to the complete posterior distribution. Estimations of the random variable θ

$\{\boldsymbol{\theta}\}$ may be obtained via one of several estimators, such as the mean or maximum of the posterior distribution. Such distributions are useful for assuming patch-wise shapes of images and clusters, for example. In the case of image representation, each Gaussian may be tilted, expanded, and warped according to the covariance matrices $\boldsymbol{\Sigma}_i$. One Gaussian distribution of the set is fit to each patch (usually of size 8×8 pixels) in the image. Notably, any distribution of points around a cluster (see k-means) may be accurately given enough Gaussian components, but scarcely over $K=20$ components are needed to accurately model a given image distribution or cluster of data.

4.

$$p(\boldsymbol{\theta} | \mathbf{x}) = \sum_{i=1}^K \phi_i \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \quad \{\displaystyle p(\boldsymbol{\theta} | \mathbf{x}) = \sum_{i=1}^K \tilde{\phi}_i \mathcal{N}(\tilde{\boldsymbol{\mu}}_i, \tilde{\boldsymbol{\Sigma}}_i)\}$$

5.

A prior distribution is placed over the parameters describing the topic distributions, using a Dirichlet distribution with a concentration parameter that is set significantly below 1, so as to encourage sparse distributions (where only a small number of words have significantly non-zero probabilities). Some sort of additional constraint is placed over the topic identities of words, to take advantage of natural clustering. For example, a Markov chain could be placed on the topic identities (i.e., the latent variables specifying the mixture component of each observation), corresponding to the fact that nearby words belong to similar topics. (This results in a hidden Markov model, specifically one where a prior distribution is placed over state transitions that favors transitions that stay in the same state.) Another possibility is the latent Dirichlet allocation model, which divides up the words into D different documents and assumes that in each document only a small number of topics occur with any frequency.

6.

Parameter estimation and system identification Parametric mixture models are often used when we know the distribution Y and we can sample from X , but we would like to determine the α_i and θ_i values. Such situations can arise in studies in which we sample from a population that is composed of several distinct subpopulations. It is common to think of probability mixture modeling as a missing data problem. One way to understand this is to assume that the data points under consideration have "membership" in one of the distributions we are using to model the data. When we start, this membership is unknown, or missing. The job of estimation is to devise appropriate parameters for the model functions we choose, with the connection to the data points being represented as their membership in the individual model distributions. A variety of approaches to the problem of mixture decomposition have been proposed, many of which focus on maximum likelihood methods such as expectation maximization (EM) or maximum a posteriori estimation (MAP). Generally these methods consider separately the questions of system identification and parameter estimation; methods to determine the number and functional form of components within a mixture are distinguished from methods to estimate the corresponding parameter values. Some notable departures are the graphical methods as outlined in Tarter and Lock and more recently minimum message length (MML) techniques such as Figueiredo and Jain and to some extent the moment matching pattern analysis routines suggested by McWilliam and Loh (2009).

7.

Handwriting recognition The following example is based on an example in Christopher M. Bishop, Pattern Recognition and Machine Learning. Imagine that we are given an $N \times N$ black-and-white image that is known to be a scan of a hand-written digit between 0 and 9, but we don't know which digit is written. We can create a mixture model with $K = 10$ different components, where each component is a vector of size N^2 of Bernoulli distributions (one per pixel). Such a model can be trained with the expectation-maximization algorithm on an unlabeled set of hand-written digits, and will effectively cluster the images according to the digit being written. The same model could then be used to recognize the digit of another image simply by holding the parameters constant, computing the probability of the new image for each possible digit (a trivial calculation), and returning the digit that generated the highest probability.